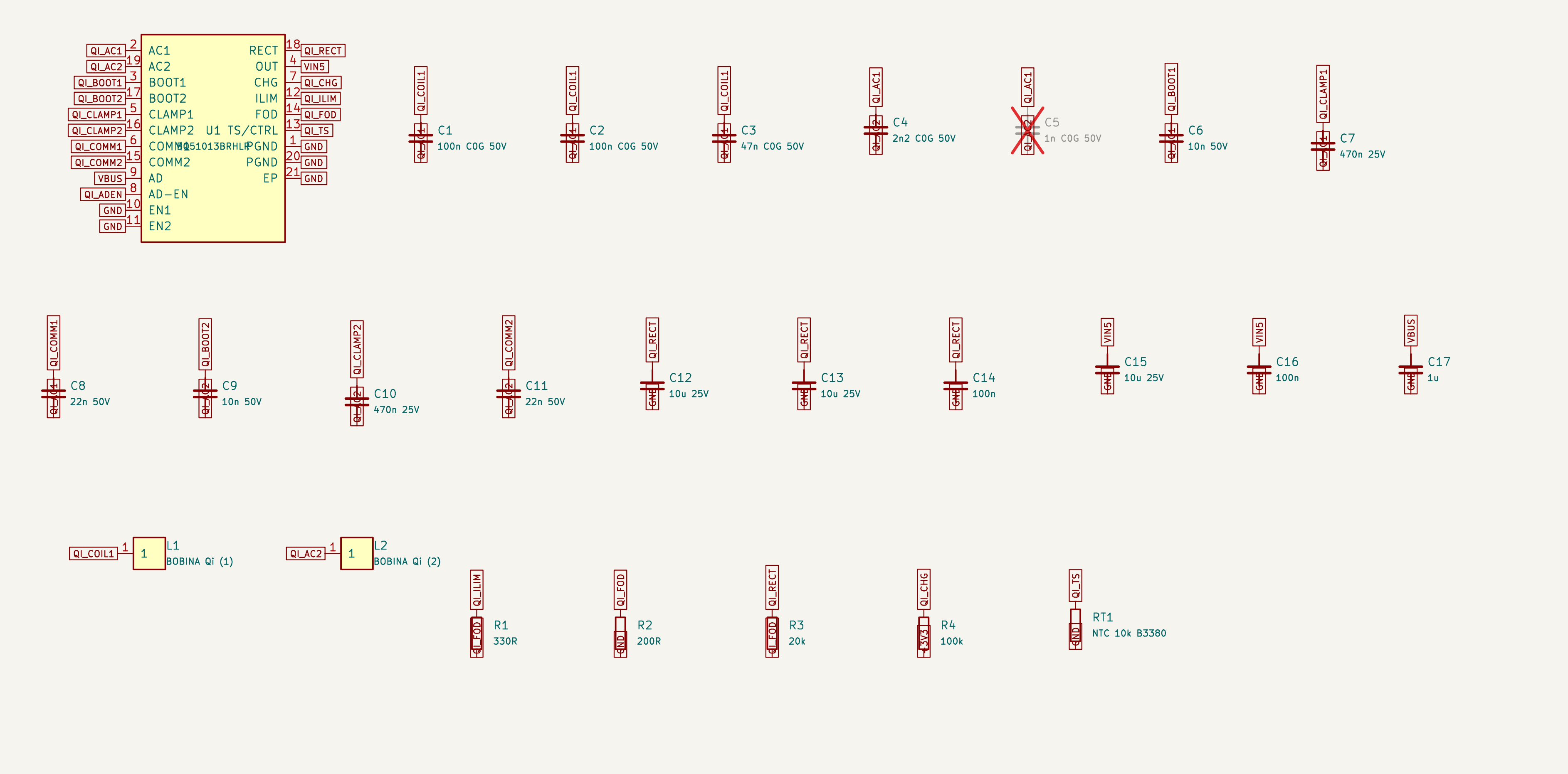
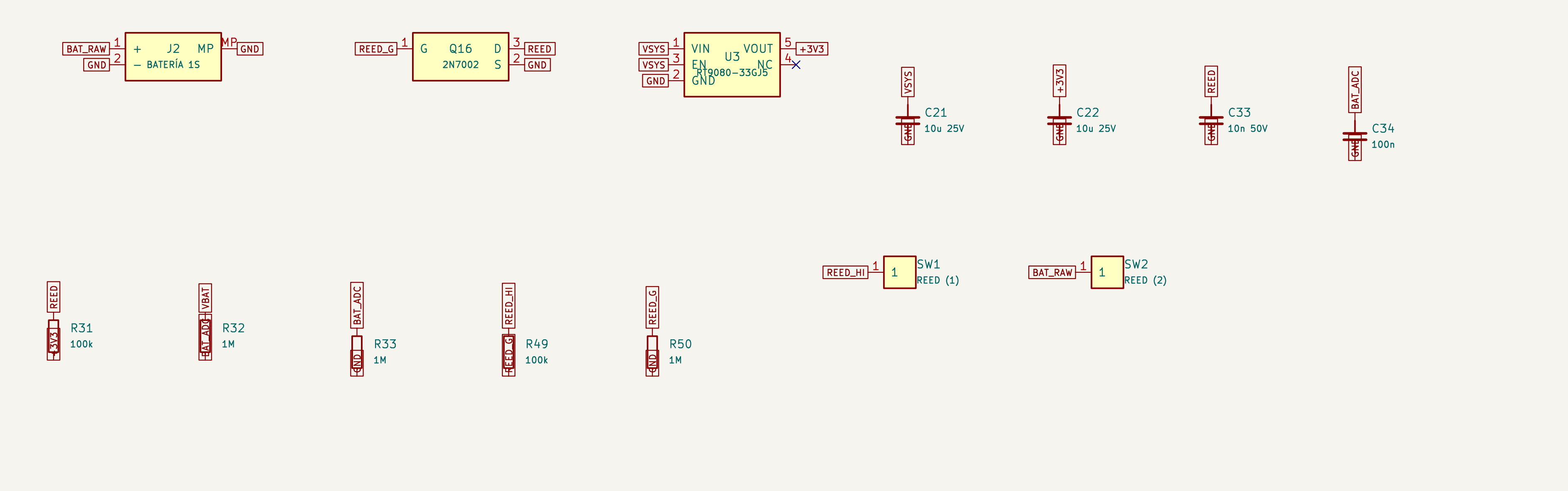
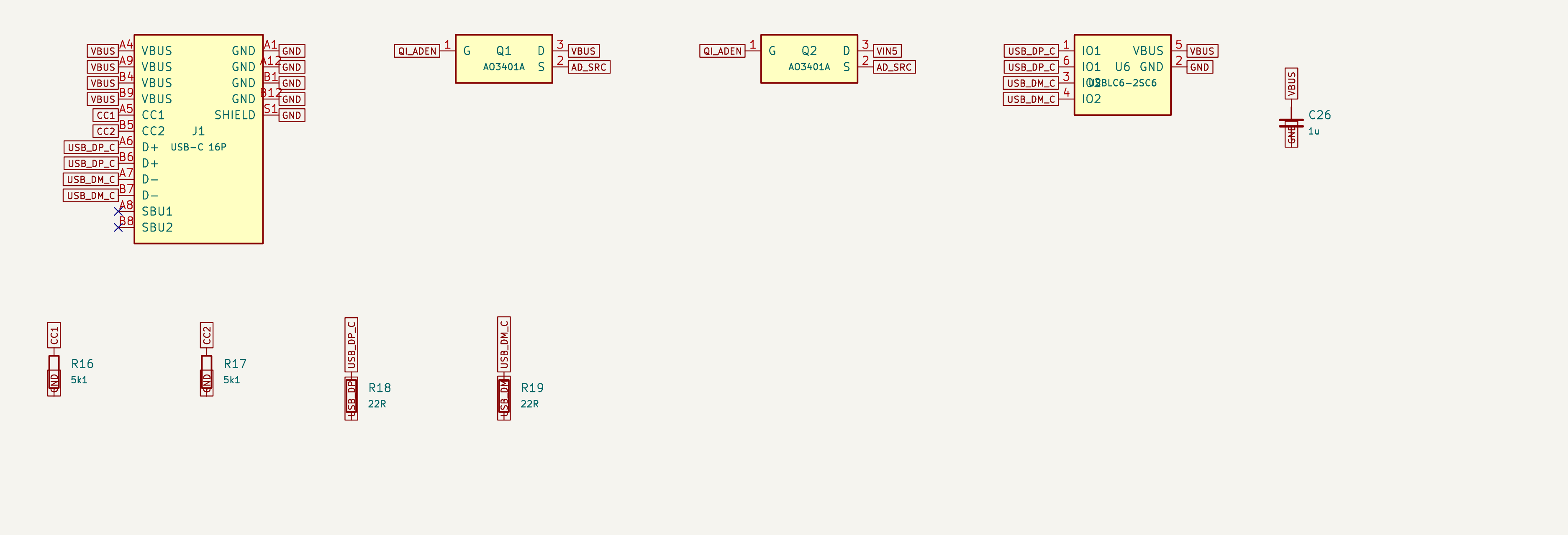
## RECEPTOR Qi (BQ51013B) · bobina Ø25 · USB por AD/AD-EN


U1 BQ51013BRHLR
AC1 AC2 BOOT1 BOOT2 CLAMP1 CLAMP2 COMM1 COMM2 AD AD-EN EN1 EN2
RECT OUT CHG ILIM FOD TS/CTRL PGND PGND EP
QI_AC1 QI_AC2 QI_BOOT1 QI_BOOT2 QI_CLAMP1 QI_CLAMP2 QI_COMM1 QI_COMM2 VBUS QI_ADEN GND GND
QI_RECT VIN5 QI_CHG QI_ILIM QI_FOD QI_TS GND GND GND
C1 100n C0G 50V
C2 100n C0G 50V
C3 47n C0G 50V
C4 2n2 C0G 50V
C5 1n C0G 50V
C6 10n 50V
C7 470n 25V
C8 22n 50V
C9 10n 50V
C10 470n 25V
C11 22n 50V
C12 10u 25V
C13 10u 25V
C14 100n
C15 10u 25V
C16 100n
C17 1u
L1 BOBINA Qi (1)
L2 BOBINA Qi (2)
R1 330R
R2 200R
R3 20k
R4 100k
RT1 NTC 10k B3380

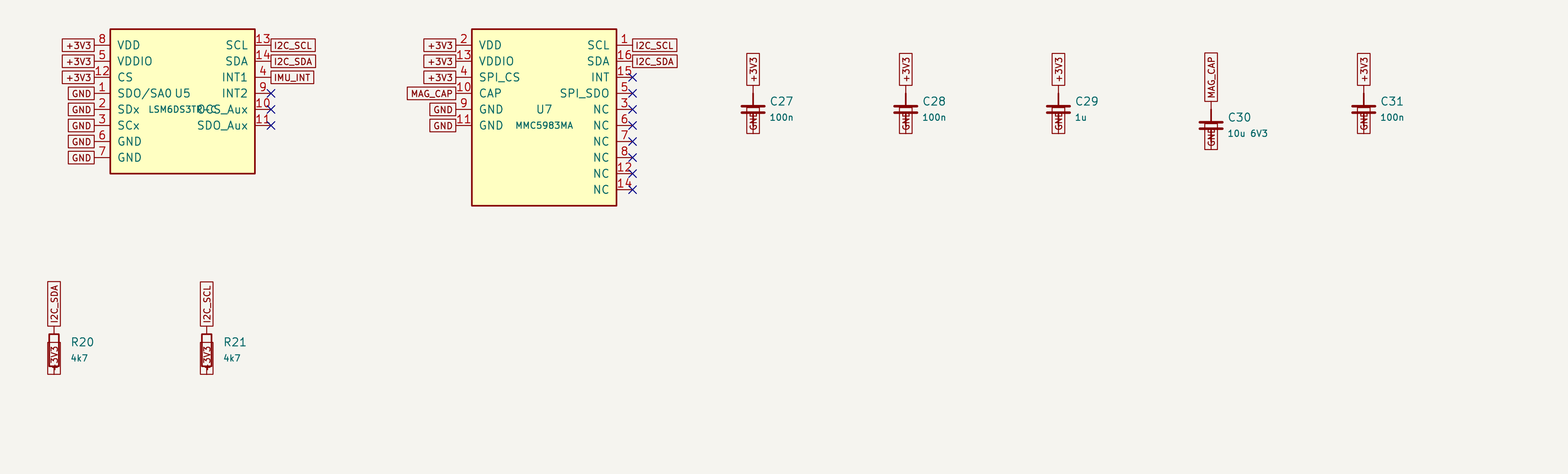
## 3,3 V (RT9080) · REED · BATERÍA (ADC)


J2 BATERÍA 1S
BAT_RAW GND MP GND
Q16 2N7002
REED_G REED GND
U3 RT9080-33GJ5
VIN EN GND VOUT NC
VSYS VSYS GND +3V3
C21 10u 25V
C22 10u 25V
C33 10n 50V
C34 100n
SW1 REED (1)
REED_HI
SW2 REED (2)
BAT_RAW
R31 100k
R32 1M
R33 1M
R49 100k
R50 1M

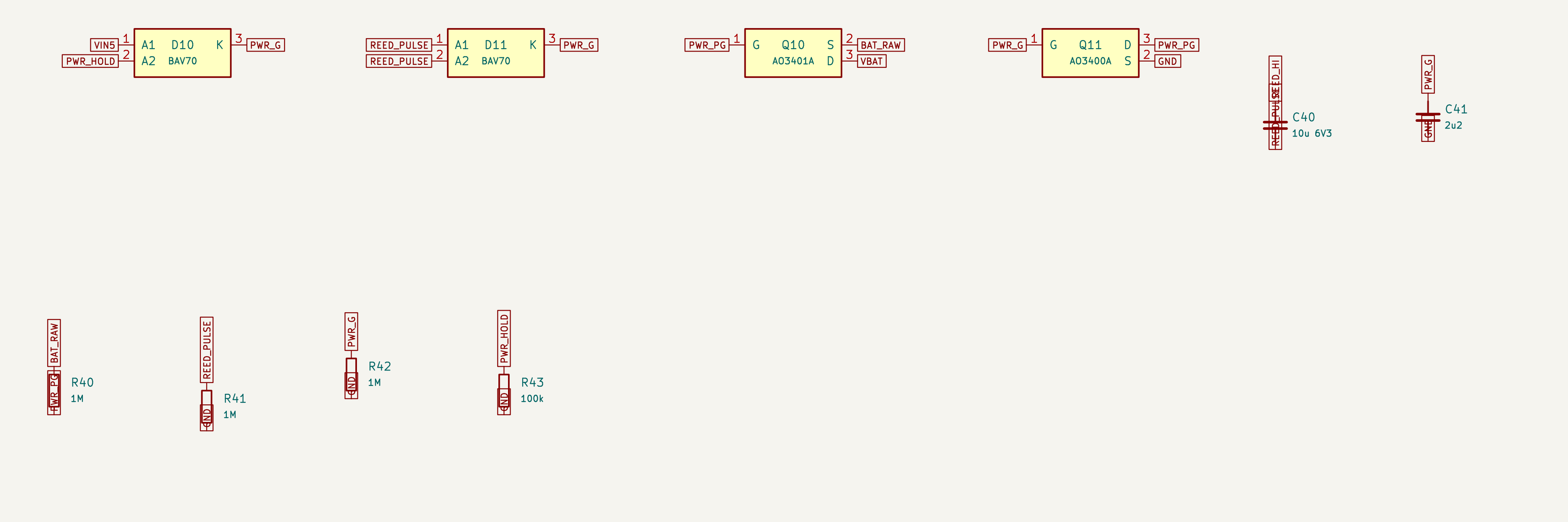
## USB-C (programación) · ESD · entrada AD


J1 USB-C 16P
VBUS VBUS VBUS VBUS CC1 CC2 D+ D+ D− D− SBU1 SBU2
GND GND GND GND SHIELD
USB_DP_C USB_DM_C CC1 CC2
Q1 AO3401A
QI_ADEN VBUS AD_SRC
Q2 AO3401A
QI_ADEN VIN5 AD_SRC
U6 USBLC6-2SC6
IO1 IO1 IO2 IO2 VBUS GND
C26 1u
R16 5k1
R17 5k1
R18 22R
R19 22R

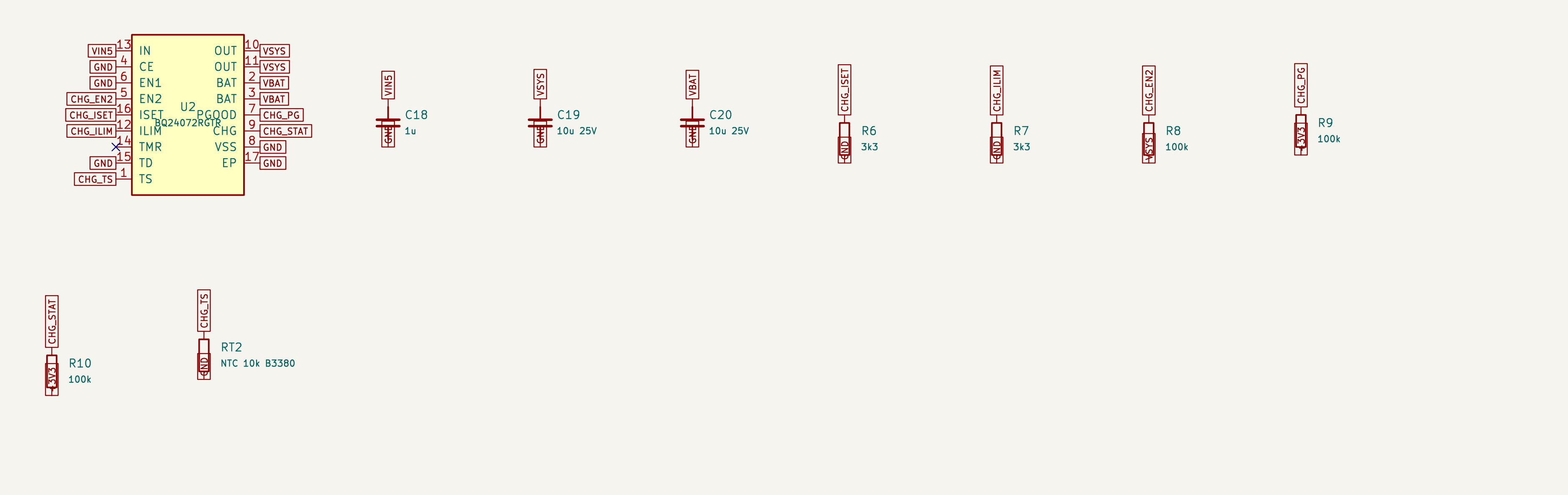
## IMU LSM6DS3TR-C + MMC5983MA (I²C)


U5 LSM6DS3TR-C
VDD VDDIO CS SDO/SA0 SDx SCx GND GND
SCL SDA INT1 INT2 OCS_Aux SDO_Aux
I2C_SCL I2C_SDA IMU_INT
U7 MMC5983MA
VDD VDDIO SPI_CS CAP GND GND
SCL SDA INT SPI_SDO NC
MAG_CAP
C27 100n
C28 100n
C29 1u
C30 10u 6V3
C31 100n
R20 4k7
R21 4k7

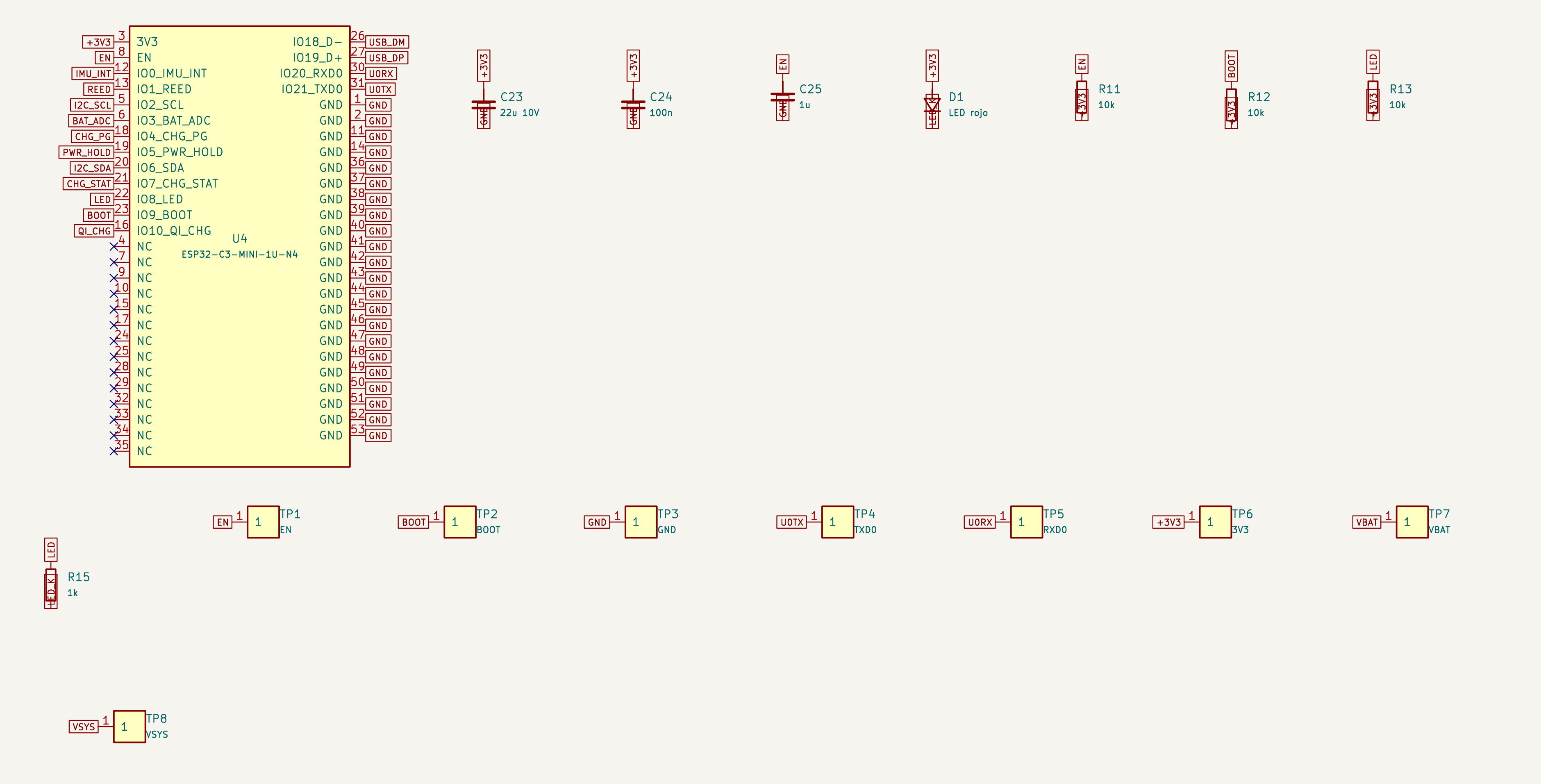
## CORTE DE BATERÍA: enclavamiento con MOSFET (reed · 5 V · PWR_HOLD)


D10 BAV70
VIN5 PWR_HOLD PWR_G
D11 BAV70
REED_PULSE REED_PULSE PWR_G
Q10 AO3401A
PWR_PG BAT_RAW VBAT
Q11 AO3400A
PWR_G PWR_PG GND
C40 10u 6V3
C41 2u2
R40 1M
R41 1M
R42 1M
R43 100k

## CARGADOR CON CAMINO DE POTENCIA (BQ24072) · batería 1S 500 mAh


U2 BQ24072RGTR
IN CE EN1 EN2 ISET ILIM TMR TD TS
OUT OUT BAT BAT PGOOD CHG VSS EP
VIN5 GND GND CHG_EN2 CHG_ISET CHG_ILIM GND CHG_TS
VSYS VSYS VBAT VBAT CHG_PG CHG_STAT GND GND
C18 1u
C19 10u 25V
C20 10u 25V
R6 3k3
R7 3k3
R8 100k
R9 100k
R10 100k
RT2 NTC 10k B3380

## ESP32-C3-MINI-1U-N4 · arranque · LED · pruebas


U4 ESP32-C3-MINI-1U-N4
3V3 EN IO0_IMU_INT IO1_REED IO2_SCL IO3_BAT_ADC IO4_CHG_PG IO5_PWR_HOLD IO6_SDA IO7_CHG_STAT IO8_LED IO9_BOOT IO10_QI_CHG NC
IO18_D− IO19_D+ IO20_RXD0 IO21_TXD0 GND
USB_DM USB_DP U0RX U0TX
C23 22u 10V
C24 100n
C25 1u
D1 LED rojo
R11 10k
R12 10k
R13 10k
R15 1k
TP1 EN
TP2 BOOT
TP3 GND
TP4 TXD0
TP5 RXD0
TP6 3V3
TP7 VBAT
TP8 VSYS

Generado por fuente/avo_girth_pcb.py (mismo origen de datos que la placa). Las conexiones son etiquetas globales: dos pines con la misma etiqueta están unidos.

**AVO Performance Horse**
Sheet: /
File: AVO_GIRTH_PCB.kicad_sch
**Title: AVO GIRTH · placa de la cincha · REV B**
Size: User | Date: 2026-10-29 | **Rev: B**
KiCad E.D.A. 9.0.9 | Id: 1/1